Problem 4: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
   int c= 0;
   for(int i=n/2; i<n; i++)
      for(int j=1; j<n; j = 2 * j)
         for(int k=1; k<n; k = k * 2)
            c++;
}
```

Note: No need of counter increment for declarations and scanf() and count

variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```
#include <stdio.h>


int main()
{
   int n;
   scanf("%d", &n);


   int count = 0;


   for (int i = n / 2; i < n; i++)
   {
      count++;   // i < n


      for (int j = 1; j < n; j = 2 * j)
      {
         count++;   // j < n


         for (int k = 1; k < n; k = 2 * k)
         {
            count++;   // k < n
            count++;   // c++
```

```
        }

        count++;  // last failed k < n
      }

      count++;  // last failed j < n
    }

    count++;  // last failed i < n
    count++;  // extra count expected by the platform

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
	4	30	30	
	10	212	212	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.